

BAKHMUT, Ukraine

WMO: 345100

Lat: 48.5572N

Lon: 38.0147E

Elev: 124

StdP: 99.84

Time Zone: 2.00 (EUE)

Period: 84-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.0	-17.7	-23.5	0.5	-20.5	-20.4	0.6	-17.4	7.0	0.5	6.2	-1.6	1.1	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	32.4	20.5	30.6	19.6	28.9	18.9	21.5	29.4	20.6	28.4	19.9	27.1	2.3	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.1	14.1	24.3	18.1	13.2	23.3	17.3	12.5	22.6	63.1	29.5	60.0	28.3	57.3	27.1	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.5	5.4	4.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.8	-3.8	-3.3	1.7	9.6	16.5	19.4	21.1	20.5	15.1	8.4	1.9	-2.5
	DBStd	10.43	6.18	6.83	5.06	3.86	3.44	3.10	2.86	3.25	4.28	5.11	5.88	5.58
	HDD10.0	1852	429	374	258	54	1	0	0	0	7	92	248	389
	HDD18.3	3761	687	607	516	262	78	25	7	14	114	308	494	647
	CDD10.0	1402	0	1	1	42	202	281	344	326	159	42	3	1
	CDD18.3	268	0	0	0	0	21	56	93	81	17	0	0	0
	CDH23.3	2988	0	0	0	5	212	507	1026	991	243	4	0	0
	CDH26.7	997	0	0	0	0	31	141	368	384	73	0	0	0
Wind	WSAvg	1.9	2.1	2.0	2.3	2.3	1.8	1.7	1.6	1.5	1.6	1.8	1.8	2.1
Precipitation	PrecAvg	801	91	75	70	54	61	74	67	45	52	52	68	94
	PrecMax	1087	174	182	145	123	138	145	155	87	144	109	135	178
	PrecMin	628	32	19	33	9	7	15	9	7	4	19	8	37
	PrecStd	126	37	33	28	25	27	34	33	22	34	21	34	40
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.5	13.5	17.4	24.0	29.3	31.7	34.6	35.2	33.6	24.3	15.5	11.5
		MCWB	5.5	9.5	9.8	14.4	17.9	20.0	21.1	20.9	20.1	15.3	12.2	11.0
	2%	DB	5.3	8.6	13.6	21.0	27.5	30.0	32.1	32.7	28.1	21.6	13.1	7.1
		MCWB	4.5	6.8	8.0	13.2	16.9	19.5	21.1	20.2	17.4	14.1	11.2	6.1
	5%	DB	4.2	5.8	10.5	19.1	26.1	28.4	30.4	30.7	26.0	18.8	10.9	5.3
		MCWB	3.4	4.3	6.9	11.6	16.8	19.5	20.2	19.0	16.8	12.9	9.2	4.4
	10%	DB	2.6	4.0	8.3	17.1	24.0	26.3	28.4	28.8	23.6	16.1	8.9	3.2
		MCWB	1.9	2.9	5.5	10.6	16.2	18.7	19.0	18.6	15.7	12.3	7.3	2.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.0	9.6	10.2	15.2	19.6	22.5	23.4	21.8	20.5	16.5	12.7	11.0
		MCDB	6.2	12.0	15.3	22.3	25.6	28.2	30.6	31.1	30.6	22.3	14.5	11.6
	2%	WB	4.7	6.9	8.5	13.7	18.8	21.1	21.9	20.9	18.9	15.3	11.2	6.3
		MCDB	5.2	8.2	12.4	20.0	24.8	27.3	30.2	29.8	26.6	19.2	13.2	7.0
	5%	WB	3.4	4.4	7.3	12.6	17.9	20.2	20.7	20.1	17.3	13.9	9.4	4.5
		MCDB	4.1	5.4	10.4	17.3	23.5	26.4	28.7	28.5	23.7	17.4	10.6	5.3
	10%	WB	2.0	3.0	5.7	11.3	17.0	19.3	19.8	19.3	16.2	12.6	7.4	2.5
		MCDB	2.6	4.1	7.9	15.7	22.7	25.0	26.7	27.1	22.3	15.9	8.4	3.1
Mean Daily Temperature Range	5% DB	MDBR	5.2	6.9	7.5	10.4	11.8	11.3	12.0	13.3	12.5	10.2	6.0	4.9
		MCDBR	5.1	8.5	12.1	13.8	14.2	13.8	14.9	16.6	17.0	14.5	8.1	5.0
		MCWBR	4.5	6.6	7.6	7.2	6.4	5.8	6.0	6.6	7.7	8.0	5.4	4.5
	5% WB	MCDBR	4.6	7.2	11.0	11.7	11.7	11.6	13.7	14.4	14.5	11.8	7.0	4.8
		MCWBR	4.4	5.8	7.3	6.8	5.6	5.6	6.1	6.2	7.5	7.3	5.3	4.7
Clear-Sky Solar Irradiance	taub	0.331	0.362	0.395	0.460	0.421	0.418	0.445	0.443	0.414	0.378	0.354	0.334	
	taud	2.170	2.099	2.142	2.047	2.244	2.283	2.216	2.206	2.250	2.315	2.288	2.225	
	Ebn at Noon	696	760	804	787	842	846	815	797	780	745	672	640	
	Edh at Noon	83	112	127	155	132	128	135	131	114	90	75	70	
All-Sky Solar Radiation	RadAvg	0.97	1.91	2.98	4.34	5.80	6.23	6.13	5.56	3.86	2.23	1.02	0.63	
	RadStd	0.21	0.39	0.45	0.45	0.50	0.54	0.40	0.39	0.41	0.34	0.20	0.10	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (8 neighbors)	+0.70	N/A	N/A	+1.09	N/A	N/A	N/A	-180	+121	+79

Nomenclature: See separate page